

Benjamin Stahl

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EDUCATION

Lehigh University, Bethlehem, PA — *B.S. Computer Science & Business, Minor in Data Science*

GPA: 3.52 | Expected Dec 2026 | Study Abroad: Barcelona, Spain (Summer 2025)

Relevant Coursework: Computer Architecture, Systems Software, Operating Systems, Machine Learning, Data Structures & Algorithms, Database Systems

TECHNICAL SKILLS

- **Languages:** C/C++, SystemVerilog, Python, Perl, Bash, Tcl, Java, Go, JavaScript, Assembly (x86)
- **Hardware/DV:** UVM, functional coverage (covergroups, crosses, URG), SVA, DPI (C++↔SV), regression triage, VCS, Coverity
- **Tools:** Git, SVN, Verdi, Linux, LSF, CMake/Ninja, Docker, SQL, AWS, Jira, CI/CD

WORK EXPERIENCE

Cisco, Maynard, MA | ASIC Design Verification Engineer Intern May 2026 - August 2026

- Root-caused a false-failing C++ verification checker by reconstructing simulation timelines against RTL sampling logic; proved the DUT correct and redesigned the checker around a convergence model, eliminating false failures while preserving detection of stuck-output, wrong-settled-value, and cross-channel-leakage bugs
- Designed a SystemVerilog functional-coverage model for a MACsec (IEEE 802.1AE) block, closing a false-coverage-closure gap by adding a missing physical-bandwidth dimension and its crosses; validated across the widest and narrowest legal lane topologies
- Authored a MACsec datapath reset test generator supporting two reset scopes with a DPI-based post-reset IP re-arm handshake; root-caused 3 distinct failures and added scoped assertion waivers for the reset window rather than weakening DUT checks
- Drove the verification build from 30 compiler errors across 15 files to zero (largest block ~226 errors) and re-enabled the strict build and Coverity static-analysis pass in nightly regression; fixed 5 high-severity defects
- Triaged ~40 outstanding nightly regression failures across two datapath blocks, automating reproduction directly from the results database and clearing ~1/3 of the backlog as stale with documented rationale

EarthPulse AI, Barcelona, Spain | Geospatial Data Science Intern May 2025 - July 2025

- Automated satellite data ingestion and preprocessing in Python, reducing pipeline runtime 40% and enabling daily refresh cycles
- Built and deployed pipelines processing 1TB+ of satellite imagery, generating triplet/pair training data for unsupervised super-resolution models

TECHNICAL PROJECTS

[Tclslang \(Tcl Extension for Verilog Analysis\)](#) C/C++, Slang (SystemVerilog), Tcl

- Built a C++ Tcl extension wrapping the Slang SystemVerilog frontend, exposing parsed design structure — modules, ports, cells, connections, and drivers — as first-class Tcl objects with dynamically registered commands
- Implemented hierarchical cell traversal and driver resolution, classifying each net driver as const/var/net and surfacing data type, net type, and dimension metadata for structural RTL analysis
- Packaged as a loadable shared library (CMake, RHEL) with golden-output regression checks against a Verilog test suite

[Encrypted C++ Network Server & Key-Value Store](#) C++, Linux Sockets, OpenSSL

- Engineered a crash-resilient server enforcing two-phase locking so in-memory updates and their durable log records commit atomically, preserving operation ordering under concurrent writers
- Optimized performance using low-level POSIX file I/O (fsync, write) and custom binary serialization to minimize overhead and guarantee data durability
- Secured user authentication by implementing salted SHA-256 password hashing using OpenSSL

[Chrome Recaptcha Harvester \(Open-Source\)](#) Python, Flask, ChromeDriver, Multithreading

- Maintained a popular open-source automation framework (95+ GitHub ★) for collecting and serving reCAPTCHA tokens, used by developers for scalable data-collection workflows

LEADERSHIP & INVOLVEMENT

Computer Science & Business Association, VP of Development / Technical Development Chair (2024–2026)

- Directed the **Lehigh Valley Collegiate Hackathon** (~100 students, six universities); secured **\$10K** in sponsorships